

Satvik Panchal

San Francisco | panchalsatvik@gmail.com | [linkedin.com/in/satvik-panchal](https://www.linkedin.com/in/satvik-panchal) | satvikpanchal.github.io

EDUCATION

Arizona State University

M.S. Computer Science (GPA: 3.9/4.0)

Tempe, AZ

May 2025

Arizona State University

B.S. Computer Science (GPA: 4.0/4.0)

Tempe, AZ

May 2024

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, R, C#, SQL

Backend & APIs: FastAPI, Flask, GraphQL, REST APIs, Pydantic, Docker

ML & AI: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Optuna, OpenAI API, Gemini API

Cloud & Data: AWS (EC2, ECS, Fargate, ECR, Lambda, SQS, CloudWatch, IoT Core), Azure AI Foundry, PostgreSQL, MySQL, Apache Kafka, Apache Spark, Splunk, GitHub, JIRA, Linux

PROFESSIONAL EXPERIENCE

Relma AI

Remote

Founder & AI Engineer

November 2025 – Present

- Engineered proprietary AI pipelines for structured understanding of analytics tools across **20+** enterprise platforms.
- Deployed containerized AI services on **AWS** using **ECR** and **ECS (Fargate)** behind an **Application Load Balancer**, and implemented unified long-term memory with **LangChain** and **FAISS** for cross-session LLM reasoning.

MOOG Space and Defense

Gilbert, AZ

Software Engineering Intern

May 2024 – April 2025

- Engineered an optimized end-to-end **SQL Server** database of flight-critical hardware components with a **Python Streamlit** UI, reducing retrieval time from **5 minutes** to under **3 seconds** for engineers.
- Implemented rigorous unit testing in **C** for flight-critical software for NASA's Artemis III using **Unity/CMock**, achieving **100%** code coverage and deployed onto **FPGA**-based development kits.

Solisa AI (Startup)

Chandler, AZ

Machine Learning Engineering Intern

May 2024 – July 2024

- Built and trained a **Python TensorFlow** model on over **13K** data points, achieving an AUC score of **75%** while reducing false negatives using **DBSCAN** clustering and **NearMiss** undersampling.
- Automated GPU-based hyperparameter tuning in **Python** with **Optuna**, improving prediction accuracy by **7%**.

Astro Seed

Tempe, AZ

Machine Learning Engineer Intern

August 2023 – May 2024

- Led a **5-person** team to build and fine-tune a **Python TensorFlow ResNet50 CNN** for plant nutrient deficiency detection, achieving **97%** accuracy and boosting precision by **12%**.
- Earned **2nd** place out of **130+** teams for delivering a high-performing ML solution.

Nike

Beaverton, OR

Software Engineering Intern

June 2022 – August 2022

- Reduced manual audit review time by **20%** by automating log-based audit tracking in **Splunk Cloud** using **SQL**.
- Integrated **Python**-based sentiment analysis into **Splunk** dashboards for real-time monitoring across **20+** data sources.

PERSONAL PROJECTS

AWS AI-Powered Face Recognition | Python, AWS, MTCNN, FaceNet

February 2025 – May 2025

- Architected a real-time edge AI pipeline using **AWS IoT Greengrass**, **EC2**, and **MQTT** for on-device face detection (**MTCNN**), achieving **0.77s** average latency across **100** test images.
- Built a cloud-to-edge inference backend with **AWS Lambda**, **SQS**, **FaceNet**, and a custom **Python** autoscaler, processing **100** requests in **96 seconds** with zero failures.

AI-Powered Job Tracker | Python, FastAPI, Gmail API, OAuth 2.0, HTML/CSS/JS

July 2025 – Present

- Built an AI-powered job tracking system in **Python** using **Gmail API** and **OAuth 2.0 (GCP)** with a **FastAPI** backend and **Tailwind CSS** UI, integrating a **LangChain**-based LLM pipeline with **Ollama SLMs** and **Gemini LLM** for structured metadata extraction from job emails.

EXTRACURRICULAR EXPERIENCE

Python Users Group

Tempe, AZ

President

August 2022 – May 2024

- Led a **100+** member club, organizing workshops on artificial intelligence, data engineering, and cloud computing.